Gayatri Vidya Parishad College of Engineering Autonomous

Mechanical Engineering | Course Code: ME201

Mechanics of Solids - Mechanics of Solids Stress Analysis & Beam Deflection

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Overview: Mohr circle of plane stress, Euler column buckling, torsion equation for circular sha

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Simple Stresses & Strains

Hooke law, Poisson ratio, Modulus of elasticity, Bulk modulus, Thermal stresses in compound bars, Vo

Unit 2: Shear Force & Bending Moment Diagrams

SFD and BMD for cantilevers, simply supported and overhanging beams under concentrated, UDL, and uni

Unit 3: Bending & Shear Stresses in Beams

Pure bending theory, Flexure formula sigma/y = M/I = E/R, Section modulus, Shear stress distribution

Unit 4: Torsion of Shafts & Buckling of Columns

Torsion formula T/J = tau/r = G*theta/L, Polar modulus, Power transmission in shafts, Euler buckling

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Plot SFD and BMD for a simply supported beam with a uniformly distributed load and calculate the point
2. Derive the Euler critical buckling load P_cr = pi^2 * E * I / L_eff^2 for a column with one end fixed
3. Draw Mohr Circle for an element subjected to sigma_x = 80 MPa, sigma_y = -40 MPa, and tau_xy = 30 MPa,

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* S. Timoshenko - Strength of Materials CBS Publishers
* Ferdinand P. Beer, E. Russell Johnston - Mechanics of Materials McGraw-Hill 7th Ed

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